Deepak Mallampati

APPLIED AI ENGINEER - HEALTHCARE PRODUCTION ML

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US-based, F-1 OPT

PROFESSIONAL SUMMARY

AI/ML Engineer with 4+ years shipping production Generative AI, RAG, and ML systems in regulated healthcare and enterprise settings. Built clinical RAG pipelines (dense retrieval + cross-encoder re-ranking) that materially reduced hallucination, fine-tuned LLMs via 4-bit QLoRA, and stood up RAGAS/BERTScore evaluation harnesses to catch regressions before release. Delivered privacy-preserving clinical ML (re-identification risk 3.38%→0.32% while retaining 99% predictive performance) and fault-tolerant pipelines with 60% fewer failed runs - ready to bring that rigor to a live clinical AI fleet.

CORE COMPETENCIES & SKILLS

Production ML • Clinical RAG • LLM Fine-tuning (PEFT/LoRA/QLoRA) • Evaluation Harnesses • Data Flywheel • Vertex AI / SageMaker / Azure ML • MLOps • Healthcare AI

Generative AI & LLMs: GPT-4o, GPT-5, Claude Sonnet, Gemini, LLaMA | LangChain/LangGraph | RAG Architecture | Prompt Engineering | Multi-Agent Systems | Fine-tuning (LoRA, QLoRA, PEFT) | Hugging Face Transformers

Machine Learning & NLP: PyTorch, TensorFlow, Keras | NER, Text Classification, Summarization | Model Optimization | A/B Testing | Drift Detection | RAGAS Evaluation | BERTScore

Vector Search & Retrieval: FAISS, Pinecone, Weaviate | Semantic & Hybrid Search | Cross-Encoder Re-ranking | Document Chunking | Semantic Caching

MLOps & Cloud: AWS (EC2, S3, Lambda, Bedrock, SageMaker), Azure (OpenAI, AI Services, Content Safety), GCP Vertex AI | Docker, Kubernetes | Terraform | CI/CD | MLflow, Weights & Biases | Observability

Security & Compliance: HIPAA, SOC 2, PCI DSS | OAuth 2.0, JWT, RBAC | PII Masking/Redaction | Prompt Injection Mitigation | Audit Logging | Zero-Trust

Languages & Tools: Python, JavaScript/TypeScript, Java, SQL | FastAPI, React | Git, Jira, Confluence | Agile/Scrum

PROFESSIONAL EXPERIENCE

Progress Solutions *Jul 2025 - Present*

AI Engineer

USA

- Architected production-grade **agentic LLM systems** on a conductor-subagent orchestration framework, decomposing complex objectives into context-scoped subagents - reducing token consumption by **42%** while sustaining task accuracy across multi-step workflows.
- Engineered high-performance **retrieval-augmented generation (RAG)** pipelines over large-scale healthcare document corpora, combining dense vector retrieval with cross-encoder re-ranking to lift answer relevance and materially reduce hallucination on domain-specific queries.
- Designed **privacy-preserving data workflows** for sensitive clinical datasets (K-anonymity, L-diversity, differential privacy) enabling compliant analytics with zero PII exposure.
- Built fault-tolerant **automation infrastructure** with scheduled batch orchestration and exponential-backoff retry logic, reducing failed production runs by **60%**.
- Established a structured **LLM evaluation and monitoring framework** (RAGAS, BERTScore, custom domain metrics) with latency/accuracy dashboards to surface regressions before release.
- Optimized inference cost and latency through prompt compression, semantic caching, and model routing, sustaining SLA targets while reducing per-query overhead.

Vanguard
Jan 2024 - Jan 2025

AI Engineer Intern

USA

- Developed **AI agents and backend services** for Vanguard's internal AI platform across **25+ AI agents** and workflows.
- Optimized LLM prompts and evaluated **GPT-4o, Claude Sonnet, and Gemini**, improving task success rates by **27%** and informing model-selection decisions.
- Engineered safeguards and content controls for secure, policy-compliant AI responses, reducing unsafe outputs by **42%**.
- Built **12 APIs and microservices**, reducing p95 latency from 1.5s to **800ms** and accelerating feature integration by 40%.
- Delivered scalable LLM applications supporting **100,000+ requests daily** across 20+ internal teams.

Kent State University
Oct 2023 - Jan 2024

ML & Gen AI Research Assistant

USA

- Built a **privacy-preserving ML pipeline** on clinical hospital-readmission data, reducing record-linkage re-identification risk from **3.38% to 0.32%** while retaining **99% of baseline predictive performance**.
- Fine-tuned **Qwen3-27B via 4-bit QLoRA** on an NVIDIA H100 with a curated instruction dataset for controllable long-form generation.

Emerson
Jun 2022 - Apr 2023

ML Trainee

India

- Developed supervised **regression and classification models** for equipment-failure prediction, anomaly detection, and capacity forecasting.
- Fine-tuned **BERT and RoBERTa** for entity extraction and text classification, achieving **89% F1-score** on custom NER tasks.

PROJECTS

MangaLens
Shipped Chrome extension (live on the Chrome Web Store) delivering real-time AI translation of manga and webtoons across 29+ sites.

career-ops
Autonomous, AI-driven job-search pipeline that scans listings, scores fit, and generates tailored applications overnight without supervision.

Privacy-Preserving Clinical ML Pipeline
Framework combining k-anonymity, l-diversity, and differential privacy to analyze sensitive patient data without exposing identities, quantifying the privacy-utility trade-off.

Story Director LLM
Fine-tuned language model generating structured short-form video narratives from a single prompt, paired with an automated rendering pipeline.

EDUCATION

M.S. in Computer Science - *Kent State University, OH* *2025*

GPA: 3.70 / 4.0